When can a prediction replace a physical trial? A production-referenced evaluation of design-stage decision rules in deep drawing

Hüseyin Tayyer Canseven[1*]

[1*]Department of Electrical Engineering, LUT School of Energy Systems, LUT University, Yliopistonkatu 34, Lappeenranta, 53850, Finland.

Corresponding author(s). E-mail(s): huseyin.canseven@lut.fi;

Abstract

Simulation and machine learning inform design-stage manufacturability decisions, yet they are judged by prediction error, not by whether their verdicts could replace a physical trial. This paper proposes a production-referenced evaluation of design-stage decision rules. Its unit, the within-run floor, is the difference that drift and scatter of production and measurement produce between two batches of one design within a run. For each evidence relation of a new design to produced ones (a sibling, a new process setting or a new family), a rule that returns meets, fails or trial is characterised by its decisive distance, beyond which at least 95 % of its verdicts are correct, and its safe distance, beyond which at most 5 % are wrong; a guard band turns them into single verdicts. An exploratory application to one laboratory deep-drawing process, with alternatives produced once and an uncalibrated simulation dataset, scored simulation-based, learned and production-record rules. The nearest produced setting, one rule for every relation, decided at 3.4 floors with a sibling and 8.5 for a new process setting. No simulation-based or learned rule was more decisive; a Gaussian process specified for small data reduced to a force trend. Borrowing the other family's errors on the same relation, not learning, made the production record safe for new settings. For a new family only simulation-based rules decided, none closer than 30 floors. Within a family, a production record indexed by its relation to the coming design is the baseline a learned manufacturability check has to beat.

Keywords: decision support, design for manufacturing, machine learning, uncertainty quantification, selective prediction, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming the decision rests on a finite-element simulation (Banabic, 2010): the team compares the predicted characteristics with the requirements and releases the alternative, changes it or sends it to a physical trial. Artificial intelligence makes this loop faster: surrogates replace simulations, models learn manufacturability from earlier and synthetic designs (Yan et al, 2026; Ghadai et al, 2018; Ballegeer and Benoit, 2026), and predictive-quality models and virtual metrology estimate quality from process data (Tercan and Meisen, 2022; Dreyfus et al, 2022). Such models are judged by their prediction error on held-out data, which leaves open the question the team has to answer: *can this prediction be relied on to decide, or is a trial still needed?* Mechanical design cannot be verified by simulation the way integrated circuits are (Whitney, 1996), and the answer depends on how much production has been seen and how the new design relates to it.

Uncertainty quantification supplies intervals (Nemani et al, 2023), and virtual metrology attaches a reliance index to each prediction (Cheng et al, 2008), but neither says how close to a requirement a verdict can be relied on, in a unit that production defines, for each relation of a new design to the produced evidence. This paper asks three questions. How can the fitness of a design-stage prediction to decide be expressed on a production-referenced scale, for any predictor and characteristic (RQ1)? How does it depend on the amount of production evidence and on how the new design relates to it (RQ2)? And, with six to nine produced alternatives, what do a generic simulation dataset, Gaussian processes, gradient boosting and calibrated intervals contribute against a production-record baseline (RQ3)?

The proposed framework has the *within-run floor* as its unit: the difference that drift and scatter of production and measurement produce between two batches of one design within a run. A rule that turns the design-stage evidence into a verdict (*meets*, *fails* or a physical *trial*) is scored, for each *evidence relation* between a new design and the produced alternatives, by two points on its operating characteristic, its *decisive* and *safe distances*; a procedure turns them into a guard band for single verdicts and prices them by the costs of errors and trials. The decisions in scope are those of process planning, the approval of process changes in series production, where a customer may require a resubmission run, and the qualification of AI-based tools that support them, taken by a team that has produced variants of a design family; a new family stands for the feasibility question before the tryout of a new die, the tryout itself, whose tool changes between loops, being out of scope.

The application uses open data: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design–process alternatives of 500 consecutive parts each on one laboratory tool set (Baum and Heinzelmann, 2026), with finite-element simulations of the same process from a generic dataset (Baum and Heinzelmann, 2025). Each alternative was produced once, and the simulations were not calibrated to the process. The study is exploratory: it shows whether the framework discriminates between rules and evidence situations on real data, and its findings are hypotheses for confirmation. The contributions are the framework (Section 3); an evaluation design that separates evidence relations, isolates what the simulation adds and tests the learners against

a production-record baseline (Section 4); findings for this process (Section 5); and a proposed reporting checklist with an open scoring function (Section 6.4).

2 Background and related work

2.1 Machine learning for manufacturability, predictive quality and virtual metrology

Design for manufacturing codifies what a process can make into rules and cost models (Boothroyd et al, 2010); machine learning learns manufacturability from CAD models (Ghadai et al, 2018; Zhang et al, 2018), synthetic sheet-metal datasets (Ballegeer and Benoit, 2026), process data and simulated design spaces (Yan et al, 2026). Predictive-quality models and virtual metrology estimate quality from process data (Wuest et al, 2016; Tercan and Meisen, 2022; Dreyfus et al, 2022), with uncertainty from Bayesian, ensemble and conformal methods (Nemani et al, 2023; Heddoub et al, 2026); automatic virtual metrology attaches to every prediction a reliance index and a similarity index that flags unfamiliar process data (Cheng et al, 2008, 2012), the closest analogue to the guard band of this paper. Models are carried across products and processes by transfer learning and domain adaptation (Maschler and Weyrich, 2021), also from simulation to real parts (Tercan et al, 2018); in sheet forming they predict defects and springback from simulated and measured data (Cao et al, 2024; Heinzelmann et al, 2025b), and discrepancies between synthetic and real deep-drawing data are being addressed (Baum et al, 2026b). These models, and language models in case studies (Makatura et al, 2024) and benchmarks (Picard et al, 2025), are evaluated by held-out accuracy or coverage, not by their decisions against production.

2.2 Simulation credibility and the validation of forming simulation

Verification and validation ask whether a model is accurate enough for its intended use (Oberkampf and Roy, 2010), and credibility assessment scales the required evidence with the consequence of a wrong decision (ASME, 2018). Bayesian calibration estimates calibration parameters together with a model discrepancy (Kennedy and O'Hagan, 2001; Higdon et al, 2004), without which they are biased and over-confident (Brynjarsdóttir and O'Hagan, 2014; Arendt et al, 2012); multi-fidelity models combine a cheap and an expensive model through a fitted scale and a discrepancy (Kennedy and O'Hagan, 2000; Le Gratiet and Garnier, 2014). In sheet forming, friction (Hol et al, 2012), the unloading and hardening behaviour behind springback (Yoshida and Uemori, 2002; Wagoner et al, 2013) and tool deflection (Pilthammar et al, 2018) are the main sources of simulation error; robustness analysis propagates input scatter (Wiebenga et al, 2014; Chen et al, 2022), draw-in is measured and controlled in line (Havinga et al, 2020; Fischer et al, 2018), simulation datasets support surrogate and transfer learning (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a), and the simulation-to-part deviation has been decomposed statistically (Baum et al, 2026a). How close to a requirement a calibrated or learned prediction can decide remains open.

2.3 Decisions with abstention and their evaluation

In conformity assessment a measured value proves conformance, proves non-conformance or leaves the decision open, with guard bands set by the measurement uncertainty (ISO, 2017) and global consumer's and producer's risks that follow from a prior distribution of the measurand (Joint Committee for Guides in Metrology, 2012). The operating characteristics of acceptance sampling and selection give the probability of a decision against the true quality, with an indifference zone inside which no decision is required (Bechhofer, 1954). A classifier with a reject option trades errors against abstentions (Chow, 1970; Geifman and El-Yaniv, 2017), as does regression with a reject option (Zaoui et al, 2020), and learning to defer passes difficult cases to an expert (Madras et al, 2018). Cost-sensitive evaluation (Elkan, 2001) and decision curves (Vickers and Elkin, 2006) price the errors; appropriate reliance on automation requires that its reliability be visible (Lee and See, 2004), though displayed confidence can calibrate trust without improving joint accuracy (Zhang et al, 2020, 2021). The decisive and safe distances are such operating characteristics, and the guard band is the counterpart of a conformity-assessment one, equally dependent on a prior.

2.4 Production variation, measurement and design decisions

Statistical quality control compares short- and long-term variation with tolerances through capability and performance indices (Montgomery, 2019); in an initial process study an index above 1.67 meets the acceptance criteria and one between 1.33 and 1.67 may be acceptable after a review by the customer (Automotive Industry Action Group, 2006), and time-dependent models describe drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5 % (ISO, 2025), and intermediate precision under conditions that differ in time (ISO, 2023); measurement systems analysis separates repeatability from bias, linearity and stability (Automotive Industry Action Group, 2010). Design decisions between alternatives with uncertain outcomes (Hazelrigg, 1998) are taken with margins (Eckert et al, 2019) or with tests whose cost is traded against that of errors (Thomke and Bell, 2001; Salado and Kannan, 2018), and what a team knows about a new alternative depends on whether it is a variant, an adaptation or an original design (Pahl et al, 2007), which the evidence relations mirror for production evidence.

Table 1 places the framework among these methods. It adopts the indifference zone of operating characteristics, the guard band of conformity assessment, the selective risk of abstaining predictors and grouped cross-validation (Roberts et al, 2017). What is new is the production-defined unit in which every rule and characteristic is scored, the qualification by evidence relation, and their combination into a procedure for design-stage decisions that any predictor can enter.

Table 1 Positioning among methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
V&V, credibility assessment (Oberkampf and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O'Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage	yes
Virtual metrology with reliance index (Cheng et al, 2008)	process data of produced items	value, reliance and similarity index	on held-out production	flags
Conformity assessment, OC curves (ISO, 2017; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks	on the measured item	yes
Selective classification and regression (Chow, 1970; Zaoui et al, 2020)	labelled data	value or reject; risk–coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This paper	simulations, produced alternatives	verdict; distances in floors by evidence relation	yes, per verdict	yes

3 The framework

3.1 The decision problem, its scope and its loss

A design alternative a is a combination of product geometry and process settings produced as a series of parts, over which a quality characteristic c varies. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet: alternative a is *adequate* for c when the p-quantile of its parts satisfies $q_p(a, c) \leq R$. At the design stage q_p is unknown. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_p and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \tag{1}$$

The three verdicts mirror the decision rules of conformity assessment (ISO, 2017). A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of simulations of a and of the production record of other alternatives, the *calibration alternatives*. The verdict costs c_{FA} when a failing alternative is accepted, c_{FR} when an adequate

one is rejected and c_{T} when it is sent to trial, so a rule's expected cost per verdict is

$$C = c_{\mathrm{FA}} P_{\mathrm{FA}} + c_{\mathrm{FR}} P_{\mathrm{FR}} + c_{\mathrm{T}} P_{\mathrm{T}}, \tag{2}$$

which the two distances of Section 3.4 summarise at a fixed share of errors. Three limits of scope apply: the demonstration uses upper limits on the 95th percentile ($p = 0.95$, for a normal characteristic a performance index of about 0.55), lower limits following by a change of sign; verdicts are per characteristic, a design-level verdict over all characteristics being reported as an extension (Section 5.7); and attribute outcomes such as splits and wrinkles, which need a requirement on a rate, are not covered.

3.2 The within-run floor

Production evidence comes as series of consecutive parts, each cut into batches of B consecutive parts summarised by a robust centre m (here the 10 % trimmed mean). The within-run floor of characteristic c, the floor for short, is

$$F_c = Q_\alpha\big\{\, |m_i - m_j| : i < j \text{ batches of the same alternative} \,\big\}, \tag{3}$$

the α-quantile of the difference between two batches of one design. Under a stationary normal model with a standard deviation σ_{b} between batch means and σ_{w} between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_{\mathrm{b}}^2 + \sigma_{\mathrm{w}}^2/B}$ at $\alpha = 0.95$, the form of the repeatability limit (ISO, 2025, 1994); because it pools batch pairs up to 450 parts apart and grows with their lag, it is closer to a time-different intermediate-precision limit (ISO, 2023) of batch centres within a run. The reference protocol is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$; every characteristic is scored in the floor of its own geometry, and the pooled floor only converts distances to physical units. It rests on three assumptions: batches of one design differ only by drift and scatter within a run (A1); future runs vary within themselves as the runs it is estimated from do (A2); and the measuring system is repeatable, its resolution small against the floor (A3).

Four properties matter for its use. It has the unit of the characteristic, so distances in floors put all characteristics and predictors on a common scale, though equal distances do not mean equal nonconforming fractions. It is a property of a protocol, and σ_{b} and σ_{w} are reported (Online Resource 1, Table S4) so that it can be recomputed. It contains whatever drifts between batches, process and measuring system alike: without a reference artefact scanned through the series it is a production-and-measurement floor, and not the shop floor. And it excludes the variation between runs, which makes it a lower bound of the variation of series production. A new family, whose floor is unknown before it is produced, is scored in the floor of the produced family. Two alternatives a and b are resolvable for c when their effect-to-scatter ratio $\mathrm{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b|/F_c$ is at least one, $\bar{m}$ being the centre of all parts of an alternative; for a continuous parameter x with local sensitivity $s_c = \partial\bar{m}/\partial x$ the minimum resolvable change is $F_c/|s_c|$.

3.3 Decision rules

Eight rules carry the main results (Table 2). The nominal simulation (M0) is the simulation at nominal sheet thickness and friction. The scaled simulation (M1s) fits $q_p = a + b\,s_0$ to the calibration alternatives by least squares, the scale term of a multi-fidelity correction without its discrepancy. The envelope rule (M2) adds the median offset of the calibration alternatives and their largest absolute residual to the top of the simulation envelope over the dataset's thicknesses and friction coefficients. The GP-corrected envelope (M5) models the offset between q_p and the envelope as a Gaussian process (Rasmussen and Williams, 2006) fitted by maximum marginal likelihood, its noise variance bounded below by the variance between the calibration series of one geometry and force (the noise-variance bound). The nearest produced setting (NN) is the mean q_p of the calibration alternatives at the nearest force, a capability-database look-up, and the force trend (M1sn) the least-squares line of q_p in force, the counterpart of M1s without the simulation, which within a geometry is a function of the force.

The trend GP (M7n) is a learner specified for six to nine produced alternatives: a Gaussian process of q_p with a linear mean in force and lubrication whose coefficients are integrated out (universal kriging), weakly informative priors on length scales and noise instead of hard bounds, and a noise term made of the known sampling variance of each q_p and an estimated between-run nugget that absorbs no modelled effect. Its counterpart with the simulation (M7) is the autoregressive multi-fidelity model (Kennedy and O'Hagan, 2000), with the simulation in the mean and a GP discrepancy; with two calibration forces its scale on the simulation acts as a force slope. Finally, intervals are separated from predictors: P+RM puts around the prediction of a rule P a relation-matched interval, the 90 % split-conformal quantile (Lei et al, 2018) of P's absolute errors in floors over the cases of the same relation in the other family, a group-conditional construction (Vovk et al, 2005). The errors are borrowed because a family's own calibration set lacks the relation. The 21 (interpolated), 42 (extrapolated) or 63 cases behind a quantile come from one family and from one or a few shared fits, so they are not exchangeable with the other family's cases and the interval carries no coverage guarantee (Barber et al, 2023); its 90 % is nominal. No interval from n exchangeable cases promises a miss rate below $1/(n+1)$: a guaranteed 5 % needs 19 produced alternatives in a relation. NN+RM is NN with such an interval.

Besides M7, ten further rules complete the comparison. M1 adds the median offset of the calibration alternatives to M0, and M1n is their median q_p; M2n and M5n are M2 and M5 without the simulation. Mc fits one friction coefficient per lubrication pattern to the draw-ins by least squares, without a discrepancy, and Mcd calibrates the friction with a GP discrepancy (Kennedy and O'Hagan, 2001) and predicts by the posterior mixture over the friction grid. M3 corrects the matched simulation of every part by gradient boosting (Friedman, 2001), and M3n models the characteristic by boosting without the simulation, both with jackknife+ intervals (Barber et al, 2021); M0w is the range of the simulation over thickness and friction; and M6, a robustness variant, puts a conformal margin on standardised residuals around the centre of M5.

Table 2 The rules of the main results; the further rules are defined in Section 3.3

Rule	Prediction of q_p	Interval (level)	Evidence
M0 nominal simulation	simulation at nominal thickness and friction	point	none
M1s scaled simulation†	$a + b$ M0, least squares	point	q_p
M2 envelope + residual margin	envelope top plus the median offset	largest residual $(\frac{n-1}{n+1})$	q_p
M5 GP-corrected envelope	envelope plus a GP offset, ML-II	GP (nominal 0.9)	q_p
NN nearest produced setting†	mean q_p at the nearest produced force	point	q_p
M1sn force trend†	$a + b$ force, least squares	point	q_p
M7n trend GP‡	linear mean plus GP, MAP under priors	GP (nominal 0.9)	q_p
NN+RM‡	NN	relation-matched (nominal 0.9)	q_p

Fitted on the calibration alternatives; descriptors: geometry indicator, blank-holder force and lubrication rank. Pre-specified unless marked: † added after the first results; ‡ added in the revisions for this journal. Levels: guaranteed (largest residual, n calibration alternatives: 8 with a sibling, 6 for a new setting, 9 for a new family) or nominal.

3.4 Operating characteristics: decisive and safe distances

A rule is scored by judging each produced alternative as a new design at requirements placed at a known distance from the truth,

$$R = q_p(a, c) + d\,F_c, \tag{4}$$

so that a is adequate exactly when $d \geq 0$. Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and of wrong verdicts among all feasible verdicts at $|d| = \delta$: every case gives one verdict on an adequate alternative and, where the requirement is not negative, one on a failing alternative. The *decisive distance* δ_d is the smallest δ beyond which $\kappa \geq 0.95$, and the *safe distance* δ_s the smallest δ beyond which $\omega \leq 0.05$ (Fig. 1a). In plain terms, for a rule that always decides, $\delta_\mathrm{d} = \delta_\mathrm{s}$ is close to the 90th percentile of its absolute error in floors; an interval rule trades decisiveness for safety, and between δ_s and δ_d it sends designs to trial rather than misleading. Both are points on operating characteristics over a population of design situations; δ_d is the analogue of the indifference zone of a selection procedure (Bechhofer, 1954) at a probability of correct selection of 0.95.

The distances are exact up to their 0.05-floor grid. They are order statistics, so a few verdicts far out can move them: with 126 cases, some 240 verdicts, the 95 % threshold rests on about a dozen. Distances below about one floor are reported as "<1", since production reproduces the decided quantile only to about one floor within a run (Section 5.1). The distances weigh false accepts and false rejects equally, and as δ grows, failing-side requirements become infeasible for characteristics close to zero, so the failing side carries half the weight up to a few floors and a third at about 100 floors; the safe distances of each side are therefore reported too.

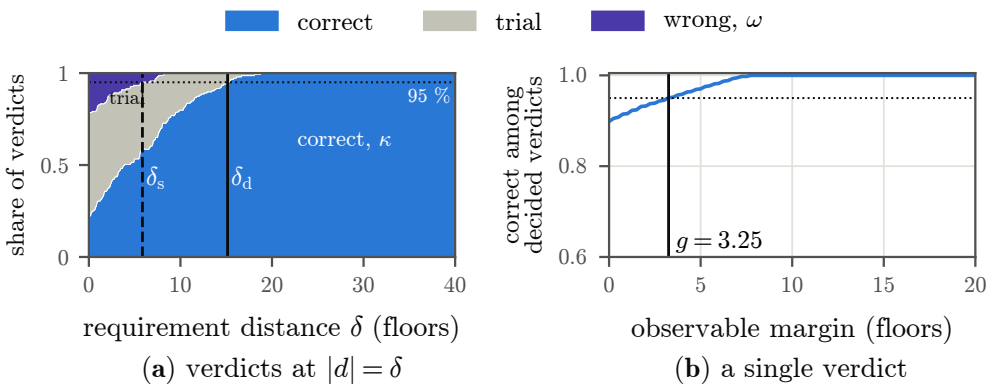


Fig. 1 Operating characteristics of one rule (M7n, extrapolated setting): verdict shares against the requirement distance, with δ_d and δ_s (a), and correct share among decided verdicts against the margin, with g (b)

3.5 Evidence relation and calibration budget

How a new design relates to the produced alternatives is part of the decision situation. With a *sibling*, an alternative at the same process setting has been produced that differs only in the factor varied at a fixed setting (here the lubrication pattern, which the nominal simulation does not represent); a *new process setting* lies between produced settings (interpolation) or outside them (extrapolation); a *new family* shares no produced alternative. A team knows, before it decides, whether the produced siblings of a new variant differ resolvably from each other, so the sibling relation is stratified by this decision-time information, not by the new design's own production. The calibration budget is the dependence of the distances on the number k of produced alternatives.

3.6 Procedure

Figure 2 turns the framework into six steps, steps 1 to 5 once per design family and repeated when production evidence is added, and step 6 for every new design. Steps 1 to 4 frame the decision (family, alternatives, characteristics, form of the requirement, costs), measure the floor with a stated protocol, state which differences production resolves, and calibrate the candidate rules with and without the simulation.

Step 5 qualifies the rules of the family by their distances for each evidence relation and fixes for each rule and relation a *guard band* g: the smallest observable margin between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts are correct. Conditional on what is observed, g depends on the distribution of requirement distances the team expects, its prior, as the global risks of conformity assessment do (Joint Committee for Guides in Metrology, 2012). The reference prior weights the 47 distances of the evaluation grid within ±20 floors equally, requirements set a few to some tens of floors from the true 95th percentile (±20 floors are about ±1 mm of mid-side draw-in, ±1.6° of wall angle, ±0.2 mm of cup depth and ±0.07–0.16 mm of dome); a release-like prior, the grid within ±5 floors, stands for requirements near capability limits, and a team replaces both by the distribution of its

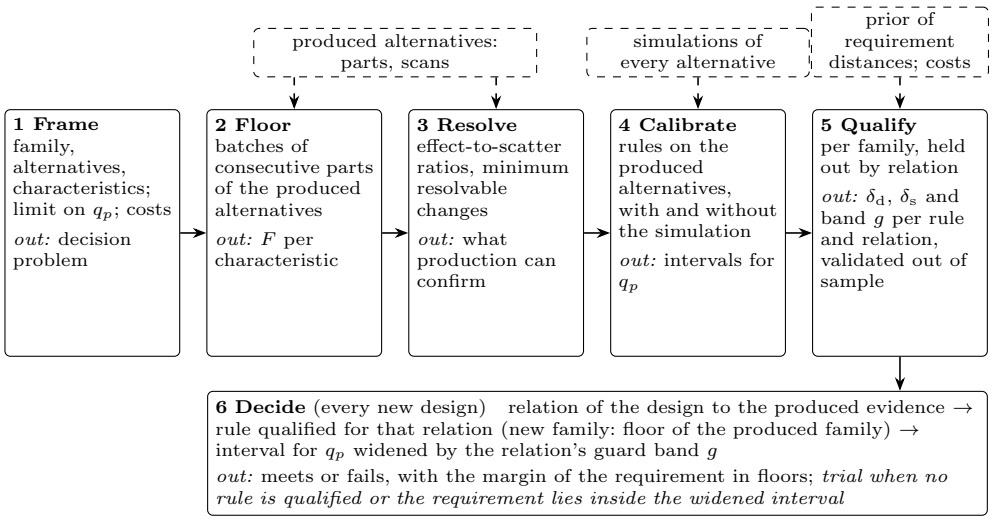


Fig. 2 The procedure: steps 1–5 once per design family, step 6 for every new design; dashed boxes are the evidence and inputs it uses

own requirements, for example the margins between past drawing limits and achieved 95th percentiles. The band is estimated per family from the relations its production exercises and validated leaving one alternative out; a relation the family's production cannot exercise (interpolation in a family produced at two forces) borrows the band of the same relation in another family, with the caveat of Section 3.3, or sends the design to trial. Stated costs may only widen the band, to the one minimising Eq. (2) over the prior, so they never lower the share correct below 95 %.

Step 6 establishes the relation of the new design, applies the rule qualified for it (with stated costs, the cheapest), widens the rule's interval by g floors and returns meets or fails with the margin in floors, or a trial when no rule is qualified or the requirement lies inside the widened interval. In pseudocode, step 5 is: for each rule and relation, collect the held-out verdicts over the prior; for $g = 0, 0.25, \ldots, 20$, compute the share correct among verdicts decided with a margin of at least g; take the smallest g beyond which it stays at or above 0.95 and, with stated costs, raise it to the g that minimises Eq. (2). Step 6 is: V = meets if $U + gF \le R$, fails if $L - gF > R$, trial otherwise.

3.7 A worked example

A process planner has to decide whether the concave cup of Section 4.1 can be run at 300 kN with medium lubrication, a setting not yet produced, without a trial run, the six alternatives at 100 and 500 kN having been produced. Step 1 frames the decision: the characteristic is the mid-side draw-in, by which the blank-holder force is set, the requirement (chosen for illustration) that 95 % of the parts draw in by at most 15.30 mm, and a trial costs a short press run, about a twentieth of a false accept. Step 2 gives the floor from the six produced series, 0.041 mm; the draw-in is quantised in

steps of 0.019 mm, three values holding a median 89 % of a series. Step 3 shows that between 100 and 500 kN a force change of about 40 kN is resolvable. Step 4 calibrates the rules on the six alternatives. Step 5 cannot qualify a rule for interpolation on this family, whose production holds no interpolated case, so it borrows from the convex family, produced at all three forces, assuming that the two families err alike: there the nearest produced setting decides at 5.3 floors and needs a band of 2.5 floors, which the stated trial cost widens to 5.0. Step 6: the nearest setting, the mean of the six alternatives at the neighbouring forces, predicts 14.99 mm, a margin of 7.4 floors, more than the band, so the planner releases the setting. The trend GP agrees with a margin of 5.4 floors; the intervals of M2, M5 and M5n contain the limit and would send the design to trial, and the nominal simulation, qualified for no relation, predicts 15.66 mm. The 500 parts had a 95th percentile of 14.88 mm, 10.0 floors below the limit; Section 5.5 reports how often such verdicts are right.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzelmann, 2026; Baum et al, 2026a). Square blanks of DP600 dual-phase steel, about 1 mm thick and taken as 210 mm wide, the blank of the simulation dataset (1 mm of width moves the measured draw-in by 0.5 mm, 10 floors), were deep drawn into cups (OP10) and cut (OP20) on an industrial press with one tool set, whose two die geometries, concave and convex side walls, differ in base curvature, wall angle and bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns applied to the blank (coarse, medium and fine, with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 g/m^2 of oil): 18 alternatives, each produced as one series of 500 consecutive parts. Every part has four load-cell signals at 300 Hz, the punch position and temperature, sheet-thickness and oil-film traverses and laser scans after both operations. The series warm up by 1.0–5.0 K over their first 150 parts from punch temperatures of 20.5–29.6 °C, their median sheet thickness is 0.986–0.993 mm, and the 100 kN series ran at 47 mm/s against 78–79 mm/s, so the extrapolation to 100 kN also changes the sliding speed and stroke rate on which the lubricant film depends. The experiments were collected between September and November 2025, in contiguous blocks of 500 per alternative; the documentation states neither the order of the blocks nor whether the parts were scanned in production order, and gives no blank size, material certificate, coil record, tool drawing, press or cushion type, or how the lubrication was applied.

4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a), a generic dataset of some 32,000 LS-DYNA simulations of the same cup family that was neither set up for nor validated against the produced process. Its documentation gives rigid tools, shell elements with seven integration points through the thickness and nodes about

1 mm apart, a scaled DP600 flow curve, constant Coulomb friction, a drawing depth of 30 mm, trimming in OP20 without modelled cutting tools, and springback after both operations; it leaves open the element formulation, whether contact accounts for the shell thickness, the yield criterion and hardening law, the blank-holder actuation and the springback procedure, and it does not represent the stroke speed. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm) and the friction coefficient (0.05–0.15) at each geometry and force, all with the nominal flow curve. A part is matched to the simulation with the nearest thickness and the friction coefficient that the dataset's linear mapping assigns to its oil film, an assumption of the dataset's authors on which M3, M3n and the matched simulations depend. The nominal simulation is at 0.99 mm, nearest the median thickness, and friction 0.10, the middle of the grid. The simulated characteristics carry extraction noise: the two directions of the quarter model differ by 1.8–2.3 floors in wall angle, and the dome is rough along the friction grid by 2.4–9.8 floors (standard deviations; Online Resource 1, Table S3).

4.3 Measurement and quality characteristics

The parts were measured with a single-sided laser line scanner, which records a height and a luminescence image of 3200 by 2000 points, 0.077 mm apart along the laser line (x) and 0.158 mm apart between lines in the direction of motion (y), with height steps of 7.75 µm, the cup bottom facing the scanner (Fig. 3). The scales are not traced to the part: the lateral one comes from the scanner specification, checked against a 9 mm wide calibration block that also gives the height (1 mm), against a nominal repeatability of 5 µm; the line spacing was estimated from the scans and corrected by a fit of the simulation's scale, which the corner draw-in, the arm angle and the dome use and the mid-side draw-in, the wall angle and the depth do not. Absolute values are therefore uncalibrated, which limits every absolute comparison with the simulation (a lateral scale error of 1 % moves the mid-side draw-in by 0.9 mm, 18 floors) but not the floor, a within-alternative difference that a constant scale error only rescales.

Seven characteristics were extracted with the same definitions from the scans and from the simulated nodes, referred to the plane of the cup bottom (Table 3): the mid-side and corner draw-in and the flange waviness, process indicators on the flange that the cut removes; the wall angle and the cup depth, measured after drawing because the cut removes the flange that defines the depth and the cut walls facing the scanner read several degrees too steep; and the arm angle and bottom dome of the cut part, the last two with requirements on absolute values. The four sides of a part measure several of them twice, which exposes direction-dependent systematic errors of up to 3.5 mm in flange width and 10° in wall angle (Table 3); the directions kept avoid the worst of them, but their residual bias is unknown. The mid-side draw-in is quantised in steps of 0.019 mm (0.38 floors), so a series takes only 6 to 21 values (Fig. 4).

4.4 Evaluation design

Every produced alternative is judged as a new design under grouped cross-validation (Roberts et al, 2017), one scheme per evidence relation. *New variant*: the alternative alone is held out and the other eight of its geometry calibrate, two siblings at its force

Table 3 The quality characteristics and their measurement by the laser scanner

Characteristic	Op.	Region; definition	Step/F	Values	Direction-dependent differences
Draw-in, mid-side	OP10	flange mid-sides; $(210 - W_x)/2$, W_x along the laser line	0.38	6–21	$W_y - W_x$ 2.29–3.47 mm; W_y, in the direction of motion, scatters 3.7–17.6 times more
Draw-in, corner	OP10	flange corners; mean of the two diagonals	<0.01	491–500	the two diagonals differ by 1.64–2.17 mm
Flange waviness	OP10	flange; residual SD about a quadratic surface	<0.01	469–494	–
Wall angle	OP10	side walls at mid-depth; mean of east and west	<0.01	497–500	convex: north wall −1.64 to −0.92° from east/west (concave −0.05 to +0.39°); south wall shadowed, 6.7–10.0° off; after the cut, north 3.3–5.6° off (convex)
Arm angle, cut	OP20	cut arms, centre lines; mean of the four arms	<0.01	496–500	–
Cup depth	OP10	flange above the bottom plane; median	<0.01	486–497	the flange plane is cut away at OP20
Bottom dome, cut	OP20	bottom sag at $r = 40$ mm; paraboloid fit	<0.01	474–497	–

Op.: operation measured (OP10 after drawing, OP20 after cutting). Step/F: median step between adjacent values in one series over the floor; Values: distinct values per series. Differences from the redundancy of the four sides, ranges of the series medians; the directions used avoid them, their residual bias is unknown. Absolute values are not calibrated against a reference at the size of the part.

among them. *New process setting*: the three alternatives of one force of a geometry are held out, the six others calibrate; 300 kN is interpolated, 100 and 500 kN extrapolated. *New family*: one geometry calibrates the rules for the other (two transfers); a few-shot variant adds one to three produced alternatives of the new family. Every calibrated rule is evaluated with and without the simulation (M1s/M1sn, M2/M2n, M3/M3n, M5/M5n, M7/M7n). Each relation has 126 cases (18 alternatives, 7 characteristics), but the units of replication are the 18 alternatives, whose characteristics are correlated.

The intervals of Table 5 resample the held-out alternatives within each geometry 1000 times (those held out together for a new setting individually), each resample with a block-bootstrap replicate of every true q_p (non-overlapping blocks of 50 parts), the fitted rules held fixed; common to all rules, they give paired intervals of differences, descriptive and not adjusted for the many comparisons. Coverage gets cluster-bootstrap intervals over alternatives where each has its own calibration set. Each truth is the 95th percentile of a single run, so a sensitivity analysis adds to every truth a between-run component of standard deviation τ floors, from near-replicate reproducibility up to the variation between series. The selection of rules and bands is cross-fitted between the geometries, chosen on one and evaluated on the other; like any cross-validated estimate, the distances describe the fitting procedure on this design rather than one fitted rule (Bates et al, 2024).

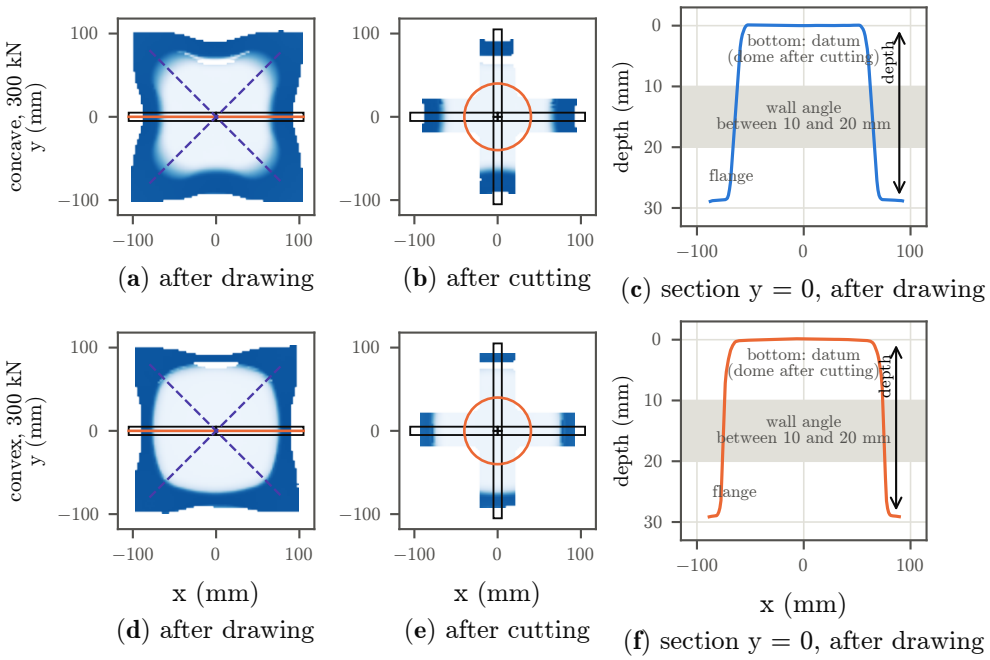


Fig. 3 Scanned parts after drawing and cutting, depth below the cup bottom (a, b, d, e), and sections (c, f); lines and boxes: directions and regions of the characteristics

4.5 Computation and development record

The constants of the floor and of the requirement grid were set before the decision results were seen, which the archive cannot verify for want of a timestamped snapshot. The rules were not fixed in advance: M0–M5 and M3n were defined first, everything else after the first results (Table 2 marks the main rules), most of it in response to reviews of earlier versions simulated with a large language model in six rounds, listed in Online Resource 1. The comparison of rules is therefore exploratory and carries the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010; Varma and Simon, 2006), which the cross-fitting estimates; every variant tried is reported here, in Online Resource 1 or in the archive (Canseven, 2026).

Computation. M5 and M5n are Gaussian-process regressions with an anisotropic squared-exponential kernel and white noise, fitted by maximum marginal likelihood on a geometry indicator, the force in 100 kN and the lubrication rank. M7n, M7 and Mcd work in pooled floors under log-normal priors with medians of 200 kN and one pattern step for the length scales and two and one floors for the GP and nugget standard deviations, Mcd with a uniform prior over the 11 friction values of the grid. M3 and M3n use histogram gradient boosting with 500 incoming conditions drawn from calibration parts of the same pattern. Software: Python 3.11, SciPy and scikit-learn (Pedregosa et al, 2011); every setting is in the archive.

Table 4 Within-run floor per characteristic, pooled and per geometry, with its spread over runs

Characteristic	F	Interval	Concave/convex	Single runs	No warm-up	F/σ_{LT}
Mid-side (mm)	0.051	0.042–0.061	0.050/0.052	0.36–1.62	0.99	1.0/1.4
Corner (mm)	0.046	0.032–0.053	0.053/0.030	0.27–1.62	0.95	1.4/1.2
Wavin. (mm)	0.0028	0.0021–0.0050	0.0037/0.0021	0.21–2.08	0.89	1.6/0.7
Wall (°)	0.078	0.054–0.10	0.081/0.078	0.30–1.74	0.80	1.3/0.7
Arm (°)	0.11	0.089–0.15	0.12/0.100	0.40–2.16	1.00	1.7/0.8
Depth (mm)	0.011	0.0090–0.012	0.0087/0.012	0.39–1.45	0.75	1.4/1.6
Dome (mm)	0.0070	0.0052–0.0084	0.0034/0.0082	0.42–1.61	0.93	1.2/1.6

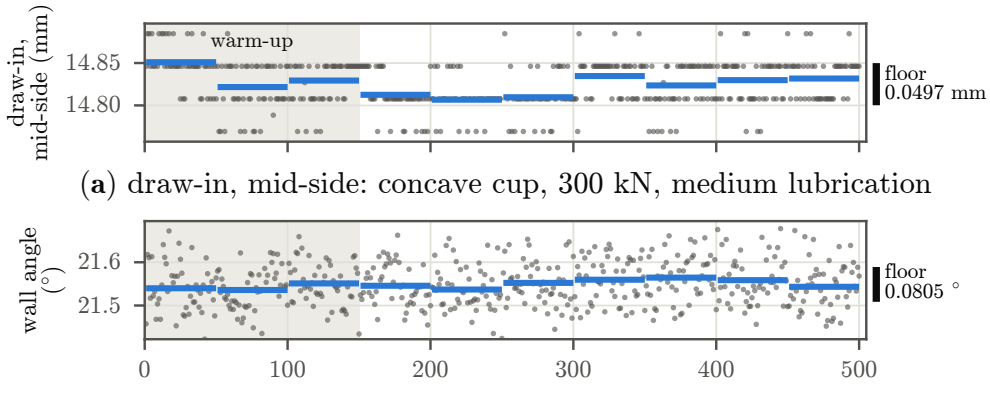
Batches of 50 consecutive parts, 95 % quantile of the difference of their 10 % trimmed means. Interval: 95 % over 2000 resamples of whole alternatives (the narrower interval over resampled batches is in the archive). Single runs: floor of one run over that of its geometry, range over the 18 runs. No warm-up: floor without the first 150 parts over the reference. σ_{LT}: standard deviation of the parts of a series (concave/convex); σ_{b}, σ_{w} and the drift ratio in Online Resource 1, Table S4.

5 Results

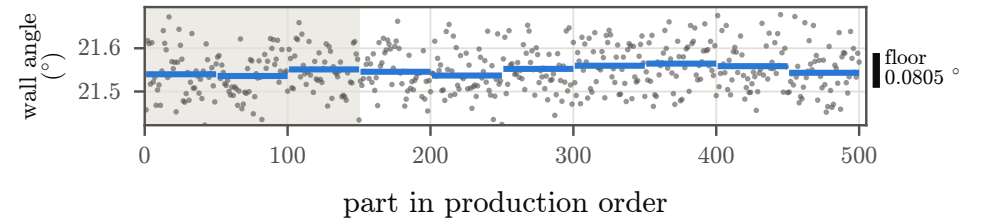
5.1 What the floor contains

Two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in and 0.078° in wall angle in 95 % of batch pairs (Table 4), so one floor is 0.7 to 1.7 long-term standard deviations of the parts. The floors are set by drift within the run (Fig. 4): without the time order they shrink by a factor of 2.1 to 4.5, and batches one position apart differ by 0.51–0.80 floors, nine apart by 1.1–1.5. Without the warm-up of the first 150 parts the floors fall to 0.75–1.00 of the reference, which keeps the warm-up because series production starts up after every break. Independent part-to-part noise, scanner repeatability included, accounts for 15–33 % of each floor, so A3 holds for repeatability and, except for the mid-side draw-in, for resolution, while bias, linearity and stability remain untested.

Whether the drift is the process or the scanner cannot be settled without a reference artefact. Its candidates are the warm-up of the tools, lubricant building up on die and blank holder, and material and blank size varying along the stack; no recorded process signal explains it (batch centres of force, punch temperature, thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20), and the batch centres of the flange width and of the cup width at 15 mm depth, which punch and die largely fix, are weakly correlated within the runs ($r = -0.27$ and $+0.27$), which bounds a common lateral scale drift of the scanner below 0.05 %, about 0.9 floor of mid-side draw-in; drift in height and of the edge detection remain untestable. Variation between runs is not in the floor: the three lubrication series of one geometry and force differ by 1.2 (waviness) to 9.8 floors (dome) at the 95th percentile of their centre differences, lubrication effects included, whereas within a run the 95th percentiles of two batches of 100 parts differ by 0.92–1.49 floors in 95 % of pairs.



(**a**) draw-in, mid-side: concave cup, 300 kN, medium lubrication



(**b**) wall angle: concave cup, 300 kN, medium lubrication

Fig. 4 The 500 parts of one run in production order with the centres of its ten batches of 50 (bars) and the floor, the warm-up shaded: mid-side draw-in (a) and wall angle (b)

5.2 What production resolves and where the simulation dataset lies

All 63 geometry contrasts exceed the floor and 105 of 126 force contrasts do, but only 51 of 126 lubrication contrasts, so a lubrication variant is often a near-replicate of its siblings. The minimum resolvable force change is 21–67 kN between 100 and 300 kN for most characteristics and exceeds either force interval for the wall angle and the dome.

The simulation dataset does not contain the produced process. The comparisons that depend on neither the assumed blank nor the scanner's scales show it: the simulated force effect on the draw-in is 2.7 to 5.1 times the measured one; the simulated wall angle of the concave cup falls by 1.5° from 100 to 500 kN, the measured one not at all; the cut arms of the concave cups bend downwards in 89–100 % of the simulations but upwards in every part, whereas the convex arm angles agree within 0.6°; and the dome ranks the geometries in reverse. The levels, which depend on blank and scales, disagree as well if both hold (Online Resource 1, Table S3): at the nominal thickness the measured centre lies within the range spanned by the friction grid in only 11 of the 42 combinations of characteristic, geometry and force; the simulated mid-side draw-in exceeds the measured one at 100 kN for every friction coefficient (by 1.8–2.0 mm at friction 0.10) and comes within 0.011 mm of it at 300 kN only at the edge of the grid on the concave cup; the corner draw-in exceeds it everywhere, by 2.6–3.5 mm at friction 0.10 and 1.3–3.3 mm at 0.15. Calibrating friction cannot repair this within the grid: all 72 within-family least-squares calibrations of Mc end at its upper edge (0.15), and with a discrepancy term the posterior of Mcd piles at the lower edge (0.05), where the simulated force effect is weakest. The dataset's mapping from oil film to friction implies 0.8–3.1 mm of mid-side draw-in per g/m^2 of oil, an order of magnitude more than the slopes within the runs and between the lubrication patterns (at most 0.09 and 0.39 mm per g/m^2), even allowing for noise in the oil readings; through it an envelope over the measured incoming variation varies with the lubrication pattern more

than production does and makes M5 less decisive with a sibling (24 against 7.2 floors). A Gaussian-process surrogate of the simulations decides as the simulation does (109 against 108 floors).

The cup depth is simulated 0.70–1.06 mm (2.5–3.7 %) deeper than measured, and the wall at 15 mm depth, set by punch and die, 2.0–2.2 mm (2.8–3.5 %) smaller in radius. The depth offset is neither definitional, bottom and flange being seen from the same side of the sheet in both, nor due to the flange inclination, which would make the measured depth larger. A drawing depth 1 mm shorter would move the wall by only 0.2–0.4 mm, and a scale error would give an offset proportional to the radius (64 and 75 mm), not 2.0–2.2 mm at both; another tool geometry fits best but is unconfirmed. Without the offset, M0 decides at 65 instead of 108 floors.

5.3 Decisions by evidence relation

Table 5 and Fig. 5 score the rules; the nominal simulation decides at 108 floors, erring mostly by false rejects.

A new variant with a produced sibling. Every calibrated rule improves on it, and the production record is the most decisive: the nearest setting, the mean q_p of the two siblings, is right beyond 3.4 floors (2.0–4.5) and the force trend beyond 3.5, while M5 and M7n decide at 6.8–7.2 floors but are safe below one floor. Stratified by what a team knows when deciding, the nearest setting needs 1.60 floors when the two produced siblings do not differ resolvably and 4.40 when they do. In physical units the nearest setting is right beyond 0.10 mm of mid-side draw-in, 0.26° of wall angle and 0.04 mm of cup depth (Table 6). One produced sibling suffices: with a single alternative at the new design's force the nearest setting decides at 3.5 floors (2.3–4.6), and further alternatives do not improve it, while they make M5 more decisive and safer.

A new process setting. With all alternatives at the new force held out, the force trend and the nearest setting decide at 8.2 and 8.5 floors and the interval rules only at 15–19, though they are safe at 5–8. Interpolating is much easier than extrapolating (the nearest setting 4.7 against 9.5 floors), but at 300 kN, the midpoint of a balanced calibration set, the nearest setting, the force trend and the centre of M5n coincide by construction, so interpolation cannot separate mean-type rules. The scaled simulation, extrapolating a fitted trend, falsely accepts failing alternatives up to 165 floors from their limit, on the cup depth, while on the mid-side draw-in and the arm angle it is more decisive than the force trend when extrapolating; the multi-fidelity model M7 fails the same way, with false accepts up to 118 floors on the cup depth (Online Resource 1, Table S2).

A new family. Scored in the floor of the produced family, as the procedure prescribes, every rule without the simulation fails (130–135 floors): nothing in one family's record describes the other. With the simulation, Mcd decides at 30 floors and M5 at 35, and M3 is safe at 17 floors and M2 at 26. Per characteristic, M5 decides at 5–10 floors for the draw-ins and the wall angle, which the forming model represents directly, but at 20–40 for depth, waviness, arm and dome. Production knowledge still carries over where a drawing nominal transfers: for the wall angle, the design angle plus the produced family's deviation from its own decides at 1.9 floors, against 7.5 for M5. Two transfers do not resolve which rule is safest.

Table 5 Decisive and safe distances (floors) by the relation of the new design to the produced evidence

	Sibling		Interpolated		Extrapolated		New family	
Rule	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR
M0	108	17/109	109	17/115	106	12/109	104	28/105
M1s†	4.9	4.7/4.9	19	30/19	19	165/7.9	121	125/118
M2	23/<1	<1/<1	18/2.7	3.7/2.4	18/9.9	8.7/10.0	41/26	27/9.7
M5	7.2/<1	<1/<1	19/2.8	2.4/3.1	19/8.9	8.4/9.7	35/31	33/15
interval of δ_d	6.6–8.4		18–20		18–20		31–39	
M7n‡	6.8/<1	<1/<1	7.4/3.4	2.2/3.4	15/5.9	7.5/<1	132/130	129/167
interval of δ_d	6.0–7.3		7.1–8.1		15–16		131–132	
NN†	3.4	2.8/3.4	4.7	3.1/4.7	9.5	11/5.6	130	129/168
interval of δ_d	2.0–4.5		4.3–5.1		6.6–11		129–131	
M1sn†	3.5	3.5/3.6	4.7	3.1/4.7	8.9	9.3/7.4	131	129/168
NN+RM‡	6.3/<1	<1/<1	9.6/<1	<1/<1	20/<1	1.5/<1	293/4.5	4.1/44
Mcd‡	7.2/<1	<1/<1	22/1.6	<1/2.5	24/5.8	6.6/2.6	30/19	29/12
Trial, first 20 parts	1.4	1.5/1.2	1.4	1.5/1.2	1.4	1.5/1.2	1.4	1.5/1.2
Trial, first 50 parts	0.90	1.0/0.85	0.90	1.0/0.85	0.90	1.0/0.85	0.90	1.0/0.85

Over all seven characteristics; one number for rules that always decide; <1: below one floor. FA/FR: safe distances on the false-accept and the false-reject side. 126 cases with a sibling and for a new family, 42 interpolated, 84 extrapolated; a new family in the produced family's floor (in its own floor, and all rules, in Online Resource 1, Table S1). Interval: 95 % over 1000 resamples of the held-out alternatives, fits fixed. Mcd: friction calibration with a discrepancy term (Section 3.3). Trial: the first n parts of the design's own run, a within-run lower bound. †, ‡: added after the first results and in the revisions for this journal.

Table 6 Decisive distance per characteristic in floors and, in brackets, in its unit

	NN			M1sn	M1s	M5	
Characteristic	Sibling	Interp.	Extrap.	Extrap.	Extrap.	New family	P_{pk}
Mid-side (mm)	2.0 (0.099)	3.8 (0.19)	6.5 (0.32)	5.7 (0.28)	4.0 (0.20)	5.5 (0.27)	1.3
Corner (mm)	4.5 (0.13)	1.9 (0.058)	6.7 (0.23)	2.8 (0.083)	2.9 (0.100)	9.9 (0.31)	2.5
Wavin. (mm)	<1 (<0.0028)	4.8 (0.014)	11 (0.040)	9.3 (0.027)	21 (0.077)	22 (0.047)	0.9
Wall (°)	3.4 (0.26)	<1 (<0.078)	1.9 (0.15)	1.8 (0.14)	2.0 (0.16)	7.5 (0.58)	1.7
Arm (°)	2.8 (0.28)	5.9 (0.59)	5.4 (0.53)	9.5 (0.94)	5.9 (0.69)	39 (3.9)	1.6
Depth (mm)	3.3 (0.039)	4.4 (0.038)	15 (0.13)	9.9 (0.086)	165 (1.4)	20 (0.20)	2.4
Dome (mm)	7.0 (0.038)	5.3 (0.044)	4.3 (0.020)	3.0 (0.020)	15 (0.056)	40 (0.21)	4.0

18 cases per characteristic with a sibling and for a new family, 6 interpolated, 12 extrapolated; a new family in the produced family's floor. In brackets: the decisive distance computed in the unit, every case scored in the pooled floor (not the distance in floors times a floor). P_{pk}: median performance index of a requirement at the decisive distance of NN with a sibling. All main rules with both distances in Online Resource 1, Table S2.

5.4 Learning given a fair test

The Gaussian processes fitted by maximum marginal likelihood reduce to the baselines in this design (Online Resource 1, Table S5). For a new setting the force length scale and noise variance of M5n sit at their bounds in most fits and its centre lies within 0.01 floor of the calibration mean in 93–95 % of the cases, with a sibling a median 0.28 floor from the nearest setting. The trend GP, specified for this regime, reduces to the

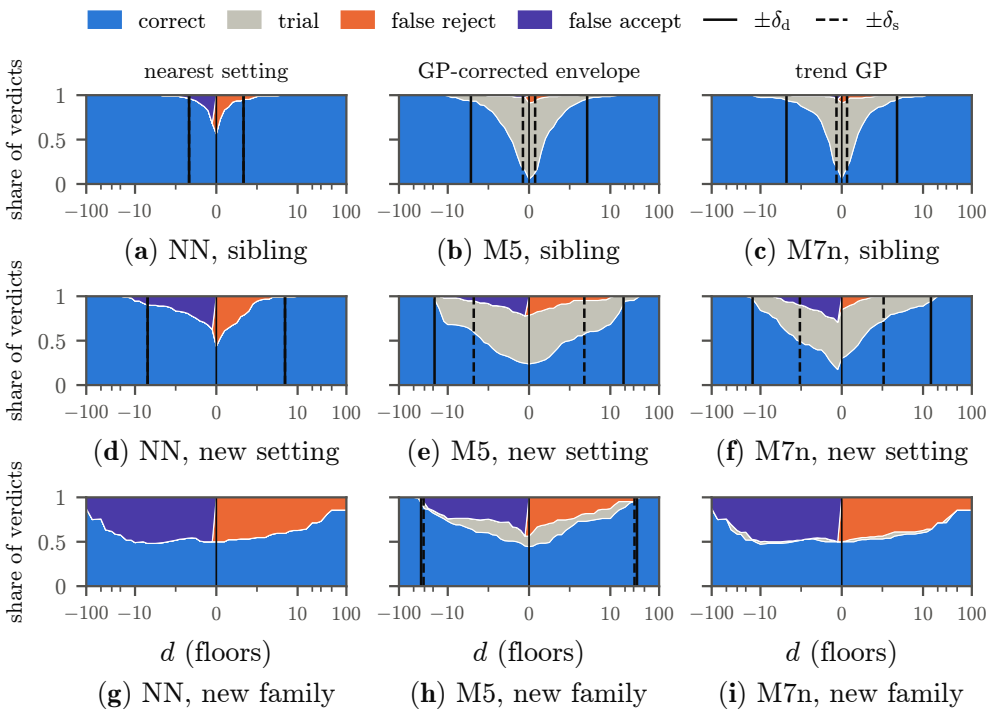


Fig. 5 Verdict shares against the requirement distance d (symmetric logarithmic axis, linear within ±10 floors; 126 cases per panel; a new family in the produced family's floor)

force trend instead: its centre lies a median 0.33–0.36 floor from it for a new setting, its force length scale stays near the prior's 200 kN, which two force levels cannot inform, and its between-run nugget, 0.4–0.5 floors, is three to four times the sampling error of the truths. It is more decisive than M5n for a new setting (14.5 against 17.2 floors) and as safe, but less decisive than the nearest setting in every relation (by 3.5 floors with a sibling and 6 for a new setting). Its intervals cover only 52 % interpolating and 56 % extrapolating (17 and 33 % on the mid-side draw-in, none on the waviness), because their width is learned from the variation between lubrication patterns at produced forces, sibling-type information; halving or doubling any of its four prior medians moves its decisive distances by at most 1.4 floors and keeps it within 0.4 floor of the force trend and its coverage for a new setting at 48–62 % (Online Resource 1, Table S6). With the simulation in its mean (M7) it gets worse for a new setting (28 floors), because the simulated force trend enters it. Tuned part-level boosting decides no closer than 11.2 floors with a sibling and 13.1 for a new setting.

Prediction and interval separate. A relation-matched interval borrowed from the other family makes even the nearest setting safe for a new setting (safe distance <1 floor, against 6.8 for M5 and 5.2 for M7n) at the decisive distance of the learned intervals (19.5 against 18.7 and 14.5 floors). The safety comes from the borrowed evidence, not from the predictor (Online Resource 1, Table S7): around the trend GP and force-trend centres the same interval decides closer when extrapolating (17.6 and

Table 7 Guard bands available when deciding (floors), with their share correct and trial rate

	Sibling			Interpolated			Extrapolated		
Rule	*g*	Correct	T	*g*	Correct	T	*g*	Correct	T
Reference prior									
NN†	0/0.5	95	2	2.5/1.5	95	18	5.75/9.5	95	58
M1sn†	0.5/1.0	95	7	2.5/1.5	95	18	6.5/5.5	95	47
M2	0	99	54	0.5/0.75	95	52	6.25/none	84	81
M5	0	99	23	0/1.75	95	57	5.0/none	83	80
M7n‡	0	98	19	1.75/0	95	28	4.75/2.75	96	60
NN+RM‡	0	99	28	0	99	42	0/0.5	97	73
Release-like prior (±5 floors)									
NN†	1.0/3.0	96	41	4.75/3.5	93	67	8.5/14.5	75	92
M1sn†	2.25/3.0	95	47	4.75/3.5	93	67	8.5/8.25	94	84
M2	0	97	76	3.25/2.25	94	85	10.25/none	72	91
M5	0/6.0	95	67	1.5/3.25	88	84	9.25/none	67	91
M7n‡	5.5/7.25	84	91	4.25/1.75	87	74	7.0/6.25	93	90
NN+RM‡	0/1.0	96	72	0.5/0	95	77	0/5.25	75	93

g: band of the family's own sibling cases with a sibling; for a new setting, which a family produced at two forces cannot exercise, the band of the same relation in the other family (concave/convex; one number when equal; none: no band up to 20 floors reaches 95 %). Correct: share correct among decided verdicts (%), with a sibling leaving one alternative out within its family, for a new setting the other family's band applied to this family. T: share of verdicts sent to trial (%). No rule qualifies for a new family. Intervals, verdict counts and the characteristics in Online Resource 1, Tables S10 and S11.

17.8 against 20.1 floors), around simulation-based centres it does not make the rule safe (false accepts up to 151–155 floors), and a trend GP calibrated on both families, borrowing through a family term, is safe at 2.75 floors. The coverage of NN+RM, 88 % interpolating and 93 % extrapolating, comes from two family swaps, not from a guarantee: it ranges from 73 to 100 % between the families and from 61 to 100 % between characteristics (Online Resource 1, Table S9). Around production-record and trend centres the relation-matched intervals have the best interval scores (Gneiting and Raftery, 2007) for a new setting (Online Resource 1, Table S8). With a sibling the Gaussian-process intervals cover 83–86 % (cluster intervals 0.72–0.94), compatible with their nominal 90 %, and the jackknife+ intervals of M3 96 % (0.93–0.98) against a guarantee of 0.78. For a new family no interval covers well except at widths that send every design to trial; with one to three produced alternatives of the new family, its own nearest setting or its offset to the other family's record decides at 5.9–10.5 floors in the floor of its nine runs, which a team with so few alternatives does not have.

5.5 Guard bands

Step 5 turns the margins into guard bands as a team could estimate them when deciding (Table 7): with a sibling from the family's own sibling cases, and for a new setting from the same relation in the other family, since a family produced at two forces has no interpolation case of its own. Under the reference prior the nearest setting keeps 95 % of its decided verdicts correct in each relation and sends 2, 18 and 58 % of the verdicts to trial; M2 and M5 send 52–57 % to trial interpolating and, extrapolating, qualify only on the convex cup, whose band gives 83–84 % correct on the concave cup.

The 95 % is an average over characteristics: the nearest setting is right in 89 % of its decided verdicts on the dome with a sibling, 88 % on the waviness interpolating and 85 % on the cup depth extrapolating (Online Resource 1, Table S11). The bands are within-run: with a between-run component of $\tau = 3$ floors the pooled bands of the nearest setting grow from 0.25, 2.0 and 7.0 to 2.9, 4.0 and 8.9 floors (Online Resource 1, Table S14).

The bands belong to the prior as much as to the rules. Under the release-like prior they widen, and the nearest setting sends 41, 67 and 92 % of the verdicts to trial and keeps only 75 % of its decided verdicts correct extrapolating; with the family's own band estimated without the held-out force cell, the nearest setting and NN+RM reach 0.72 and 0.74 extrapolating, M5 and M7n 0.88 and 0.87 interpolating (Online Resource 1, Table S10). Under the reference prior no rule qualifies for a new family, as the practice of trying out every new die would have it: Mcd reaches 95 % on one transfer with a band of 18 floors, but that band gives 73 % on the other.

5.6 Costs, trials and release-level requirements

Fig. 6 prices the verdicts by Eq. (2), a false reject costing half a false accept; the following holds under both priors and every ratio of false-reject to false-accept cost (Online Resource 1, Table S13). For a new setting the nearest setting or the force trend is the cheapest rule at every trial cost. With a sibling the nearest setting is the cheapest once a trial costs a fifth of a false accept or more; below that the cheapest rules differ by thousandths of a false accept, which this design does not resolve. For a new family trying every design is the cheapest up to a twentieth of a false accept, beyond which the cheapest rule changes with the prior and the cost ratio; minimised without the 95 % target, the cost even favours the nominal simulation, with false accepts in 14 % of its verdicts. The cost-minimising bands, in sample and pooled over the families, widen as trials get cheaper, and the 95 % bands of step 5 correspond to trials costing a tenth to a quarter of a false accept; if, as assumed here, trying a new force on an existing tool costs less, cost-optimal bands are wider. A trial, priced as error-free in Fig. 6 (whose band search runs to 40 floors), is not: scored as a rule, a trial of the first 20 or 50 parts of the design's own run decides at 1.35 and 0.90 floors, and the errors of the 20-part trial cost 0.017 of a false accept per verdict (0.037 under the release-like prior), as much as the whole expected cost of the nearest setting with a sibling at a trial cost of 0.05 (0.018). Both priors place 47 % of the verdicts on failing designs.

A limit at a performance index of 1.33, the lower end of the range in which an initial process study may be acceptable after a customer review, lies a median 1.75 floors above an alternative's 95th percentile (1.03, 2.46 and 3.14 at indices of 1.0, 1.67 and 2.0), inside every attainable distance except the nearest setting's when the produced siblings do not differ resolvably (1.60 floors). Release, however, asks whether the index is at least 1.33, a decision on $\bar{m} + 4\sigma$: scored on it, the nearest setting needs 3.35 floors with a sibling and 10.55 extrapolating, so it confirms an index of 1.33 only for designs whose true within-run index is 2.1–3.3 with a sibling and 3.8–7.4 extrapolating. Nor does a short trial settle the question once runs differ: with a between-run component of 1, 2 and 3 floors, the 20-part trial decides at 2.1, 3.6 and 5.1 floors, the nearest

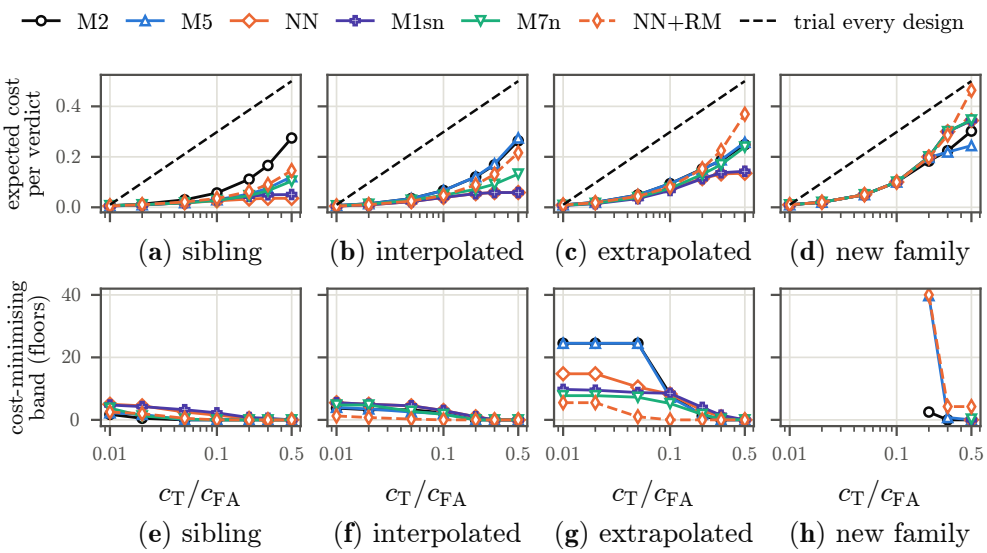


Fig. 6 Expected cost per verdict with the cost-minimising band (a–d) and that band (e–h) against the cost of a trial; reference prior, $c_{FR} = 0.5\,c_{FA}$, error-free trials, bands up to 40 floors

setting with a sibling at 3.45, 4.45 and 5.75 (Online Resource 1, Table S14). Near release level the evidence must come from replicated runs.

5.7 Sensitivity, selection and robustness

A between-run component added to every truth lengthens the distances but keeps their order: with $\tau = 3$ floors, about the largest variation between series, the nearest setting needs 5.8 floors with a sibling and 10 extrapolating, M5 9.3 and 20, and a production-record rule (nearest setting, force trend or median) is the most decisive within the family in each of 200 draws. Selected on one geometry, the most decisive rule is a production-record rule in every relation within a family and decides at 2.0–11 floors on the other; for a new family the rule selected on one transfer decides at 32–33 floors on the other. Selected for safety, the rule chosen on one geometry is safe on the other only at up to 18 floors interpolating and 11 extrapolating, and rule and band selected together keep 0.92–0.97 of the decided verdicts correct within a family, below 0.95 in four of six cases, and 0.73 for a new family (Online Resource 1, Table S15). Paired intervals between point rules span about ±2 floors with a sibling and when extrapolating, so smaller differences are not resolved by this design.

Requiring all seven characteristics of a design to meet raises the decisive distances of the production-record rules by at most 1.8 floors within a family but those of M5 by up to 16. Lower limits, truths and calibration from the first 20 or 50 parts, and further variants of data, constants and specifications leave the nearest setting within 1.2 floors of its reference distance and a production-record rule the most decisive within the family; other floor protocols rescale the distances by 0.60 to 1.42, and floors recomputed without the held-out alternative's own series move the distances of the main rules by at most 1.4 floors, that of M0 by 3.75.

6 Discussion

6.1 What the evaluation establishes

Two findings follow from the definitions and hold wherever the framework is applied: the floor puts every rule and characteristic on a common production-referenced scale, though not on a common scale of nonconforming fractions, and as a within-run quantity it bounds the variation of series production from below (RQ1); and an interval rule trades decisiveness for safety, which the two distances make explicit. The others are observations on one laboratory process, with every alternative produced once and a generic simulation dataset, and hypotheses for confirmation. The attainable distance grows with the novelty of the design: a production-record rule applied to every relation decided at 3.4 floors with a sibling, 8.5 for a new setting and 30 floors at best for a new family, and within a family the choice of rule mattered more than the relation, also among the pre-specified rules (RQ2). For this simulation dataset and for Gaussian processes and boosting fitted to six to nine alternatives, the simulation contributed little within a family, helping single characteristics when extrapolating in force, and mattered across families, for the characteristics it models directly; the learners added no decisiveness, and what made rules safe for new settings was evidence borrowed from the other family on the same relation (RQ3). Qualification by relation also exposed a danger: the scaled simulation, decisive with a sibling, accepted failing alternatives 165 floors from their limit when extrapolating, the multi-fidelity model up to 118.

6.2 Implications for practice

Within a family the production record answers most design-stage questions of a process planner or quality engineer if it is indexed by how the coming design relates to it: a sibling at the same setting settles a requirement a few floors away, a new force within the produced range almost as well, and an extrapolated force or a new family needs a trial unless the requirement is far. Under the reference prior, of 100 verdicts on new variants with a sibling the nearest setting with its band sends 2 to trial and gets 92 to 95 of the other 98 right; of 100 on extrapolated settings it sends 58 to trial, and under the release-like prior 41 and 92. And near release-level limits neither a prediction nor a short trial confirmed capability, which needed replicated runs, so the value of design-stage prediction lies in the decisions far from the limits. For a developer of an AI-based manufacturability tool the lesson is methodological: a learned model has to be compared with a production-record baseline and with its own version without the simulation, scored by evidence relation in a unit that production defines, its intervals calibrated on cases of the same relation and shown to its user with a guard band. The nearest setting is also the most transparent rule, pointing to the produced alternatives it averages, the natural default where a learned rule shows no advantage.

6.3 Simulation and machine learning in this regime

Within a design family the production record carried the decision even against a friction-fitted (censored) or rescaled simulation, but the simulation here is a generic dataset that does not contain the produced process (Section 5.2). A process-calibrated

Table 8 Guidance for design-stage manufacturability decisions, with its basis

Situation	Guidance	Basis
From the framework (prescriptive; usefulness not evaluated)		
Before relying on a predictor	Measure the floor on consecutive production with a stated protocol; report σ_{b}, σ_{w}, lag dependence and single-run floors; treat it as a lower bound of series variation.	framework
Qualifying a rule	Evaluate it by grouped cross-validation for the relation of the coming designs, per family and characteristic, siblings stratified by whether they differ.	framework
Choosing a rule	Compare it with the nearest produced setting and with its own version without the simulation, on both distances and by expected cost at the team's trial cost.	framework
Deciding a single design	Take the guard band from the relations the family's production exercises or from another family, validate it out of sample for the expected requirements and widen it where trials are cheap; a margin inside it means a trial.	framework
Observed in this study (one process, 18 alternatives produced once, 2 geometries; not general)		
New variant with a produced sibling	the nearest produced setting decided beyond 3.4 floors (1.6 when the produced siblings did not differ resolvably from each other); no learned rule was more decisive	18 alternatives
New process setting	production record 8.2–8.5 floors, interval rules 15–19; borrowing the other family's errors on the relation made it safe	6 force cells
New family	only rules with the (generic) simulation decided, at 30 floors and beyond; none qualified for single verdicts	two transfers
Release-level requirements	a limit at $P_{\mathrm{pk}} = 1.33$ lies about 2 floors above the 95th percentile; confirming the index needed replicated runs, which neither a prediction nor a short trial replaced	18 alternatives

simulation, with friction depending on pressure (Hol et al, 2012), a hardening model fit for springback (Wagoner et al, 2013) and elastic tools (Pilthammar et al, 2018), might contribute more; the per-characteristic results show where it already helps, for the draw-ins and the arm angle when extrapolating and for the characteristics it models directly in a new family. The learners failed for a reason: with six to nine alternatives at two or three forces a flexible model can identify at most a linear trend, and a well-specified one becomes that trend. Nonlinear multi-fidelity models (Perdikaris et al, 2017), physics-informed learning (Karniadakis et al, 2021), conformalised quantile regression on the parts (Romano et al, 2019) and multi-task models over more families (Bonilla et al, 2008) are the next candidates, to be scored in the same way.

6.4 Reuse for other processes and AI-based tools: a proposal

The framework needs several produced variants of a design family whose relations span those of the coming designs, a series of consecutive parts or replicated batches per variant, a design-stage predictor for every variant and characteristics measured identically on parts and predictions; here nine produced alternatives per family and 500-part series gave ten batches of 50 parts for the floor, one produced sibling sufficed for the nearest setting, and truths from the first 20 or 50 parts moved its distance with a sibling by less than half a floor. The extensions below are untested. A classifier with

Table 9 Proposed reporting checklist for evaluating a design-stage decision rule against production

Item	Why it matters	In this paper
Floor and protocol	every distance is in this unit	B, α, lag, single runs (Tables 4, S4)
Measurement	biases enter absolute comparisons	Table 3, Section 4.3
Relations, replication	alternatives, not cases, replicate	per relation, 18 alternatives
Both distances	decisiveness and safety differ	Table 5, false-accept side
Baselines	production record, no simulation	NN, M1sn; rules with and without
Intervals	coverage follows the construction	Table 2, Section 5.4
Per characteristic	pooled distances hide the worst	Table 6
Guard band, prior	the deployed part	decision-time, two priors (Table 7)
Costs and trials	deployment is a cost question	Fig. 6
Development record	selection optimism	Section 4.5, cross-fitting
Query protocol	verdicts vary between queries	language models (Section 6.4)

a fixed threshold does not take the requirement as input, so its operating characteristic is estimated by binning test cases by their true margin, which needs a continuous, production-referenced outcome for every test design that CAD-based manufacturability datasets usually lack. A language-model check is scored on a grid of requirements with repeated queries, reporting model, version, date, decoding settings and prompt paraphrases, treating refusals and hedged answers as trials and testing its verdicts for monotonicity in the requirement and for contamination by the public data. A requirement on a rate, such as the share of split parts, is scored once each alternative's rate is estimated from its series, with a floor over batches of rates; for continuous design spaces the relation becomes a distance from the produced designs in descriptor space. Table 9 is a proposed reporting checklist, modelled on model cards (Mitchell et al, 2019) and TRIPOD+AI (Collins et al, 2024); the scoring is released as a small documented function that takes intervals, truths and floors and returns distances, guard bands and trial rates (Canseven, 2026). An attempt on open powder-bed fusion data (Leirmo, 2021) failed for want of ten replicated anchors of one design per build.

6.5 Deployment

In use, a step-6 verdict would be shown with its margin in floors and in the unit of the characteristic, the relation on which it rests and the produced alternatives behind it, and a trial verdict with the requirement distance at which the rule would decide. A qualification has to be maintained: the realised errors of step-6 verdicts can be tracked against later production on a control chart of decision outcomes, which sees false accepts only, rejected designs never being made, and a rule re-qualified when its share of wrong decided verdicts exceeds 5 % beyond what the number of verdicts explains (a rise to 10 % takes about 150 decided verdicts to detect at 5 % significance and 80 % power), when the floor of new runs leaves the range of the single-run floors, or when the die is reworked. The evidence has a cost, several 500-part series per family before any rule qualifies, though one produced sibling settles most verdicts with a sibling. Whether teams decide better with the framework is an empirical question: a study with cases of known production outcome could test whether process planners

and quality engineers shown the guard-banded verdict accept fewer wrong verdicts within the safe distance than those shown the verdict alone, at an equal rate of right acceptances (Lee and See, 2004), how often they override a trial verdict, and whether the band narrows the spread of that share between teams (Zhang et al, 2021).

6.6 Threats to validity

The evaluation covers one material, one press and two geometries, with relations within a family from a 3×3 factorial in which lubrication is mostly not resolved; the procedure is general, the numbers are not. The new family rests on two transfers between geometries that differ in three features at once. Each alternative was produced once, so the reference q_p is that of one run, without the variation between days, coils and tool states; the sensitivity analysis explores its effect but cannot replace replicated runs. The floor does not separate scanner drift from process drift, and absolute values are uncalibrated. The simulation dataset was not calibrated to the process, so the findings on simulation concern this dataset. The requirements are one-sided limits on the 95th percentile of dimensional characteristics, below the conformance of series release; attribute outcomes such as splits and wrinkles are not covered; and the verdicts are marginal per characteristic, a design-level verdict being an extension. The guard band was estimated pooled over characteristics and, for new settings, borrowed between two families; its prior is a reference rather than a team's, and the costs are assumed. Rules, strata, bands and priors were partly chosen after results were seen and evaluated on the same data; the cross-fitting estimates the optimism but does not remove it, and a single analyst did the analysis, with code and data open for replication.

7 Conclusions

Design-stage manufacturability decisions can be evaluated against production on a scale production defines: the decisive and safe distances in floors tell how close to a requirement a rule decides and how close it can be relied on not to mislead, and a guard band validated for an assumed prior of requirements turns them into a rule for single decisions, priced by the costs of errors and trials. In an exploratory application to one laboratory deep-drawing process, with alternatives produced once and a generic simulation dataset, the attainable distance grew with the novelty of the design: the nearest produced setting, one rule for every relation, decided at 3.4 floors with a produced sibling and 8.5 for a new process setting, and no rule decided closer than 30 floors for a new family (the pre-specified M5 at 35). Release-level requirements lay within these distances, and confirming a performance index of 1.33 needed replicated runs, which neither a prediction nor a short trial replaced.

For design practice, the production record carried the decision within a family, which argues for indexing such records by their relation to the coming design rather than for a more elaborate model; across families only rules using the simulation carried the change of geometry. For AI-based manufacturability support, the evaluation separated what the learners were credited with: Gaussian processes and boosting fitted to six to nine alternatives reproduced the production record, and evidence borrowed from the other family on the same relation, not learning, made it safe for new settings.

The evidence is one process and two geometries, and the comparison of rules is exploratory. Four extensions follow: series from several days, coils and tool states, scanned in a documented order with a reference artefact; a process-calibrated simulation; families with continuous geometric descriptors; and a confirmatory test of the procedure, registered before the analysis, on other data. Within a design family, the baseline that a learned manufacturability check has to beat is a production record indexed by its relation to the coming design.

Supplementary information. Online Resource 1 holds the tables behind statements of the paper (Tables S1–S15), each with its result files.

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Data availability. All data are public. RDDAC: doi:10.18419/DARUS-5589 (version 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0. The extracted per-part and per-simulation feature tables and all result files are archived with the code (Canseven, 2026).

Code availability. The complete analysis pipeline, which reproduces every number in this paper from the public data, is archived on Zenodo (Canseven, 2026); `results/findings.md` maps every number to the script and result file that produce it, `results/DATA_DICTIONARY.md` translates its column and code names into the terms of the paper, and `scripts/opchar.py` is the scoring function for other datasets. A working copy is at https://github.com/htcanseven/Claude.

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author's direction to write and debug code, retrieve and check literature, run analyses and draft text. It was also used to simulate reviews of earlier versions of the manuscript in six rounds, acting as referees and editor; most rules and analyses added after the first results, listed in Online Resource 1, were added in response to these simulated reviews. None of these reviews is a review by a journal. The author designed the study, verified every analysis against the archived results, checked every reference against its published record, and takes full responsibility for the content; the language model is not an author. All figures come from the released code; none is AI-generated.

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